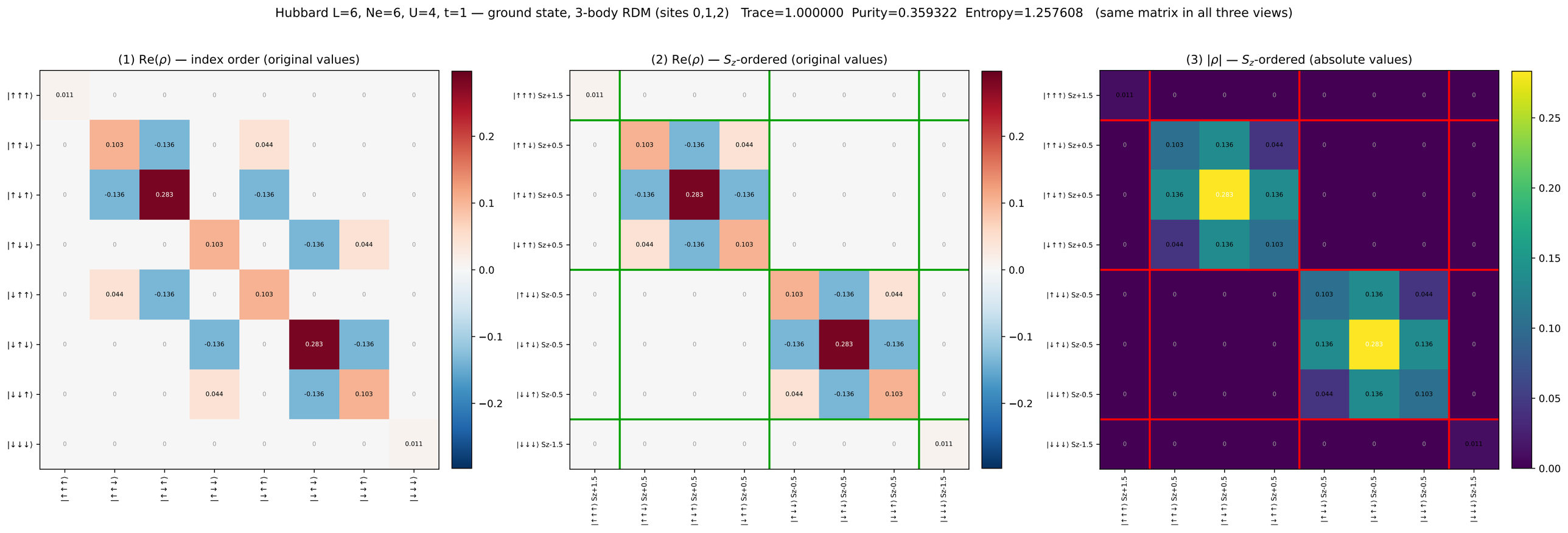
Hubbard L=6, Ne=6, U=4, t=1 — ground state, 3-body RDM (sites 0,1,2)  Trace=1.000000  Purity=0.359322  Entropy=1.257608  (same matrix in all three views)
(1) Re(ρ) — index order (original values)
(2) Re(ρ) — $S_z$-ordered (original values)
(3) |ρ| — $S_z$-ordered (absolute values)
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
|↑↑↑⟩ Sz+1.5
|↑↑↓⟩ Sz+0.5
|↑↓↑⟩ Sz+0.5
|↓↑↑⟩ Sz+0.5
|↑↓↓⟩ Sz-0.5
|↓↑↓⟩ Sz-0.5
|↓↓↑⟩ Sz-0.5
|↓↓↓⟩ Sz-1.5